

Topic: HTML

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Topics:

1. What is HTML
2. Basic Tags
3. div vs span
4. Semantic tags
5. Forms
6. Tables

HTML

HyperText Markup Language

HyperText - text which can jump to other pages

Markup Language - using tags to describe how content should appear

HTML is the building block of every web page

Invented by Tim Berners Lee in 1991

1. HTML is not case-sensitive. but prefer lowercase tags
<DIV> or <div>.
2. HTML 5 - the usage of semantic tags, multimedia inputs and improvements in forms
3. W3C - World Wide Web Consortium

Chrome - HTML Parser

Firefox - HTML Parser 2

Safari - HTML Parser 3

Edge - HTML Parser 4

<!DOCTYPE html>

lang attribute in html

1. information about the language for the document
2. it helps screen readers and assistive technologies to read content in correct language
3. It allows browser to apply correct locale based settings

DD-MM-YYYY

MM-DD-YYYY

DD-MMM-YYYY

Text block - p tag

Inline block - span tag

scope block - div tag

Defining a HTML Tag

Opening Tag:

Angle bracket open (<)

Tag Name (h1,h2,...,h6, p, div ...)

Angle bracket close (>)

Closing Tag

Angle bracket open (<)

Slash or terminator (/)

Tag name

Angle bracket close (>)

Div Tag:

1. Block level element used to group larger sections of the content
2. It takes up full width of the container, meaning it will start from a new line and stretches over the complete page

Use Cases:

1. When we need to group the content together into separate section or areas
2. For layout designing,

Soan tag:

1. Inline block element used to group small sections of text or other elements (bold or italic text).
2. It will not create a full width block. It stays within the flow of the content.

Use Cases:

1. Apply styles to small part of the text.

Semantic Tags:

1. They are used to provide meaning to the content they wrap.
2. These help search engines and browsers to understand the structure and purpose of the content
3. It also enhances the SEO (Search Engine Optimization)

SEO = ranking of the webpage on the search engine.

header: defines the header section of the document / webpage
it contains logo, navigation and introductory content

footer: defines the footer section.
it contains information for copyright, privacy policy or contact links

article: self-contained piece of content such as blog post, news article or forum post
it can stand alone and can be distributed independently.

section: defines the thematic grouping, usually with its own heading.
it can be used to group related content within article or webpage.

nav: navigation - it defines a block of navigation what e say as the menu items.

main: Represents the main content of the document.
it is used for the content which is unique to the page excluding the headers or footers or any sidebars.

figure: content like images or diagrams or charts.
figcaption: which is used to provide a caption or info about the figure or the content

strong: indicates that the text has strong importance. displayed as bold text
em: indicates the strong importance but shown as italic text

Pros of Semantic Tags:

1. SEO improvement - Search engine can understand the content better
2. Accessibility - Screen readers can navigate content more easily
3. Maintainability - Semantic code is easier for developers to read
4. Better User Experience - Clear, organized structure

Forms in HTML

1. User interaction for login, registration, payments or other information gathering

action attribute - defining the control of the form to be moved on submission
method - defining the format of the data to be passed, GET or POST

Form Input Field Types

text - can accept characters and numbers

number - can accept numbers only and also provides option to increment or decrement the values

password - it will mask the input in the format of bullets and provides an option to see the password in plain text

email - it is used to check if the input is a validate email or not. But the restriction is limited

date - adds a calendar to the field for date selection

file - adds a field to allow the user to upload files from system

Tables - used to present the tabular data

data is displayed into format of cells (row * column)

colspan - extend the occupancy of the cell to multiple columns

rowspan - extend the occupancy of the cell to multiple rows

Task: Create a skeleton for Portfolio page

1. Name, email

2. Work Experience (if any)
3. Your Education Details - display in a tabular format
4. A simple form taking input as an email